

PCC UNIT-3

1. How is Dropped Call Rate calculated? What are the key parameters involved in DCR calculation?

A. Dropped Call Rate (DCR) is calculated as the percentage of calls that are unintentionally terminated by the network after being successfully established.

The formula is:

$$DCR = \frac{\text{Number of dropped calls}}{\text{Total number of successfully established calls}} \times 100$$

Key Parameters Involved in DCR Calculation

1. Dropped Calls

- Calls that were connected but ended unexpectedly due to technical/network issues.
- Excludes calls intentionally ended by the user.

2. Total Established Calls

- The denominator includes all calls that were successfully set up and connected.
- Failed call attempts (blocked calls) are not counted here.

3. Call Setup Success Rate (CSSR)

- Ensures that only *connected calls* are considered in DCR.
- Affects the denominator indirectly.

4. Traffic Channel (TCH) Drops

- In GSM/UMTS/LTE, dropped calls are often measured as TCH drops.
- Causes include poor radio coverage, interference, hardware faults, or handoff failures.

5. Handover Failures

- If a call drops during cell handoff, it contributes to DCR.
- Mobility management parameters directly influence this.

6. Radio Conditions

- Signal strength (RxLev), signal quality (RxQual), and interference levels.
- Poor conditions increase the likelihood of dropped calls.

7. Network Parameters

- Base station hardware/software reliability.
- Transmission links, antenna alignment, and power control settings.

8. Measurement Window

- DCR is usually calculated over a defined period (e.g., per hour, per day).
- Larger windows give more stable averages.

2. Discuss the advantage of delayed handoffs with an example.

A. Delayed Handoff

- In cellular systems, a **handoff** occurs when a mobile user moves from one cell to another.
- Normally, the network triggers handoff as soon as the signal from the serving

cell drops below a threshold.

- **Delayed handoff** means the system waits a little longer before switching, even if the signal is weakening.

Advantage of Delayed Handoff

- **Reduces unnecessary handoffs (ping-pong effect):** If a user is near the boundary of two cells, the signal may fluctuate. Immediate handoff could cause repeated switching back and forth. Delaying the handoff avoids this instability.
- **Improves call continuity:** By waiting until the new cell's signal is clearly stronger, the chance of dropped calls during handoff decreases.
- **Optimizes resource usage:** Prevents wasting signaling and channel resources on frequent handoffs that aren't truly needed.

Example

Imagine a user driving along a road that passes close to the boundary of two cells:

- At point A, the signal from Cell 1 weakens slightly, and Cell 2's signal becomes stronger.
- If the system triggers an **immediate handoff**, the call moves to Cell 2.
- But as the car continues, the user re-enters Cell 1's coverage, causing another handoff back.
- This back-and-forth is called the **ping-pong effect**.

3. Explain the differences between hard handoff and soft handoff. How do they impact network performance and mobile device experience?

A.

Aspect	Hard Handoff	Soft Handoff
Definition	Break-before-make: connection with old cell is released before new one is established.	Make-before-break: mobile maintains simultaneous connections with multiple cells during transition.
Technology Use	Common in GSM, LTE (frequency-based systems).	Common in CDMA, WCDMA (code-based systems).
Process	Single link at a time; abrupt switch.	Multiple links maintained; smoother transition.
Call Drop Risk	Higher, since any delay or error during switch can drop the call.	Lower, as overlapping connections reduce risk of dropped calls.
Signaling Load	Lower, because only one connection is maintained.	Higher, due to multiple simultaneous connections.
Resource Usage	More efficient in terms of channel usage.	Less efficient, as multiple channels are reserved temporarily.
User Experience	May cause brief interruptions or dropped calls, especially at cell boundaries.	Provides seamless call continuity and better quality during mobility.
Impact on Network	Easier to manage, less complex.	More complex, requires coordination among base stations.

Key Impact

- **Hard handoff:** Simpler, resource-efficient, but riskier for call continuity.
- **Soft handoff:** More reliable for users, but increases signaling and resource overhead for the network.

4. Discuss power difference handoffs.

A. Power Difference in Handoffs

The **power difference** refers to the **difference in received signal strength (RSS)** between the serving cell and the target cell **at the moment of handoff**.

Mathematically:

$$\Delta P = P_{target} - P_{serving}$$

Where:

- P_{target} = received power from target (next) base station
- $P_{serving}$ = received power from current base station

This power difference determines **when** and **how** the handoff should occur.

The handoff must happen at the right time:

- **Too early:** The target cell may still have weaker signal → poor connection.
- **Too late:** The serving cell may lose the signal → call drop.

So, a **threshold power difference** (called *handoff margin* or *hysteresis margin*) is used.

A typical condition for handoff is:

$$P_{target} > P_{serving} + H$$

Where:

- H = **handoff margin** (usually 2–6 dB)

This means the new cell's signal must be stronger than the current one **by at least H dB** before switching.

This margin prevents “**ping-pong effect**” — rapid back-and-forth switching between two cells when signal levels are similar.

5. What is meant by Cell site hand off and Intersystem Handoff? Explain.

A. 1. Cell Site Handoff

- **Meaning:** Occurs when a mobile user moves from one cell (served by a base station) to another within the *same cellular system*.
- **Process:** The Mobile Switching Center (MSC) coordinates the transfer of the call from the current base station to the next one.
- **Purpose:** Maintains continuity of the call as the user moves across cell boundaries.
- **Features**
 - Occurs within the same system / operator / MSC
 - Frequency band and system type are usually the same
 - Relatively fast and simple
 - Requires coordination only between base stations and the same MSC
- **Example:** A user driving across town moves from Cell A to Cell B, both controlled by the same MSC. The call is handed off smoothly without interruption.

2. Intersystem Handoff

- **Meaning:** Occurs when a mobile user moves from one cellular system to another, often involving different MSCs or even different operators.
- **Process:** The call is transferred between two different systems, requiring coordination between MSCs and possibly roaming agreements.
- **Purpose:** Ensures call continuity when crossing system boundaries (e.g., moving from one city's network to another).
- **Features**
 - Involves different MSCs, possibly different frequency bands or technologies
 - Requires exchange of information between systems
 - More complex and slower than cell site handoff
 - May require authentication, frequency reallocation, or re-tuning

- **Example:** A user traveling on a highway moves out of the coverage of Operator A's system into Operator B's system. The call is handed off between the two systems.

6. What is Handoff and explain how a handoff is initiated in cellular systems.

A. Handoff (or handover) is the process of transferring an ongoing call or data session from one cell/base station to another to maintain service continuity as a mobile user moves. It is initiated when the serving cell's signal weakens and the neighboring cell's signal becomes stronger, ensuring uninterrupted connectivity.

- In cellular systems, coverage is divided into cells, each served by a base station.
- When a mobile user moves across cell boundaries, the signal from the current base station degrades.
- To avoid call drops or data interruption, the system transfers the connection to a new base station.
- This process is called **handoff** (or handover).

Handoff is Initiated

1. Signal Monitoring

- The mobile station continuously measures the signal strength and quality (e.g., RxLev, RxQual in GSM).
- These measurements are reported to the serving base station.

2. Threshold Crossing

- When the signal from the serving cell falls below a predefined threshold, and a neighboring cell's signal is stronger, a handoff condition is detected.

3. Decision Making

- The Base Station Controller (BSC) or Mobile Switching Center (MSC) decides whether a handoff is necessary.
- Decision depends on parameters like signal strength, interference, traffic load, and handoff margins.

4. Resource Allocation

- The target cell/base station is checked for available channels.
- If resources are free, the MSC/BSC reserves them for the incoming handoff.

5. Execution

- The mobile station is instructed to switch to the new frequency/channel of the target cell.
- The call or data session continues seamlessly without user interruption.

7. What is forced handoff? Explain about controlling and creating a handoff.

A. Forced Handoff

- **Definition:** A forced handoff occurs when the network compels a mobile device to switch from one cell/base station to another, regardless of whether the signal quality is still acceptable.

- **Reason:** It is usually triggered to **balance load, free resources, or avoid congestion** in the serving cell.
- **Example:** If Cell A is overloaded with traffic, the network may force some users with acceptable signal strength to hand off to Cell B to maintain overall system performance.

Controlling a Handoff

- **Network-controlled handoff:**
 - The decision is made by the network (MSC/BSC) based on signal reports from the mobile.
 - Common in GSM and LTE systems.
- **Mobile-controlled handoff:**
 - The mobile device itself decides when to switch, based on its own measurements.
 - Used in some early analog systems.
- **Mobile-assisted handoff:**
 - The mobile measures signal strength/quality and reports to the network.
 - The network makes the final decision.
 - Widely used in GSM/UMTS.

Creating a Handoff (Initiation Process)

1. **Measurement:**
 - Mobile continuously measures serving and neighboring cell signals.
2. **Reporting:**
 - Mobile sends measurement reports to the base station or controller.
3. **Decision:**
 - Network compares serving cell quality with neighbor cells.
 - If thresholds are crossed (signal below limit, neighbor stronger, or load balancing needed), a handoff is triggered.
4. **Resource Allocation:**
 - Target cell is checked for available channels.
 - Resources are reserved.
5. **Execution:**
 - Mobile is instructed to switch to the new frequency/channel.
 - Call/data session continues seamlessly.

8. Explain types of handoffs and why they needed?

A. Types of Handoffs

1. **Hard Handoff**
 - *Break-before-make:* The connection with the old base station is released before the new one is established.
 - Common in GSM and LTE.
 - Simpler, but risk of call drop is higher.
2. **Soft Handoff**
 - *Make-before-break:* The mobile maintains simultaneous connections

with multiple base stations during transition.

- Common in CDMA/WCDMA.
- Provides smoother call continuity, but uses more resources.

3. **Mobile-Controlled Handoff (MCHO)**

- The mobile device itself decides when to switch, based on its own signal measurements.
- Used in some early analog systems.

4. **Network-Controlled Handoff (NCHO)**

- The network (MSC/BSC) makes the decision based on reports from the mobile.
- Common in GSM/LTE systems.

5. **Mobile-Assisted Handoff (MAHO)**

- The mobile measures signal strength/quality and reports to the network.
- The network makes the final decision.
- Widely used in GSM/UMTS.

6. **Intra-system Handoff**

- Occurs within the same cellular system (same MSC).
- Example: moving from one cell to another in the same city.

7. **Inter-system Handoff**

- Occurs between different cellular systems (different MSCs or operators).
- Example: moving from one operator's coverage area to another.

8. **Delayed / Power Difference Handoff**

- Triggered only when the new cell's signal is significantly stronger than the current one.
- Reduces unnecessary ping-pong handoffs.

Handoffs Are Needed

- **Mobility Support:** Users move across cell boundaries while on calls or data sessions.
- **Call Continuity:** Prevents dropped calls when the serving cell's signal weakens.
- **Quality of Service:** Maintains strong signal strength and reduces interference.
- **Resource Optimization:** Balances load across cells and avoids congestion.
- **User Experience:** Provides seamless connectivity without interruptions during movement.

9. Explain briefly Mobile Assisted Handoff (MAHO) and soft handoff?

A. **Mobile Assisted Handoff (MAHO)**

- **Definition:** In MAHO, the mobile device itself measures the signal strength and quality of the serving cell and neighboring cells.
- **Process:**
 - The mobile reports these measurements to the network (BSC/MSC).
 - The network uses this information to decide whether a handoff is

needed.

- **Advantage:** Faster and more accurate handoff decisions, since the mobile is closest to the radio environment.
- **Usage:** Widely used in GSM and UMTS systems.

Soft Handoff

- **Definition:** A *make-before-break* handoff where the mobile maintains simultaneous connections with more than one base station during transition.
- **Process:**
 - The mobile communicates with multiple base stations at once.
 - The network combines signals, ensuring continuity until the old link is safely released.
- **Advantage:** Reduces call drop risk and improves quality, especially in CDMA/WCDMA systems.
- **Trade-off:** Consumes more network resources since multiple channels are reserved temporarily.

10. Discuss about how a two-level hand-off algorithm is useful in delaying a handoff.

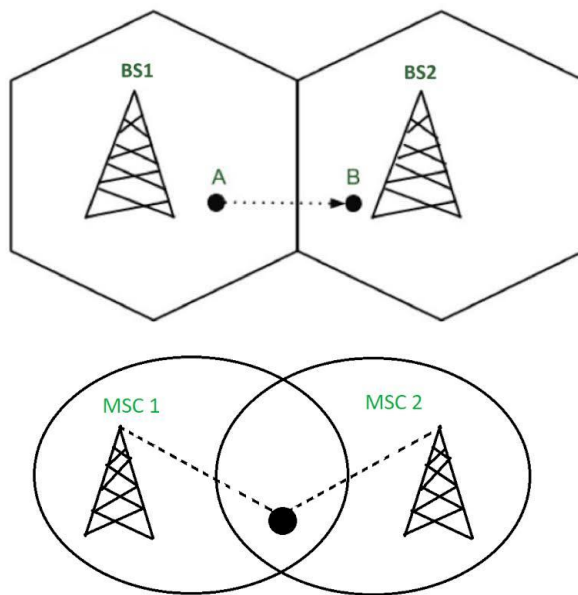
A. Two-Level Handoff Algorithm

- A **two-level handoff algorithm** introduces *two thresholds* for signal strength or quality instead of just one.
- **Level 1 (Warning Threshold):** When the serving cell's signal drops below this level, the system starts monitoring neighboring cells more closely but does not immediately trigger a handoff.
- **Level 2 (Critical Threshold):** If the serving cell's signal falls below this lower level, and a neighboring cell's signal is stronger, the handoff is executed.
- By using two thresholds, the algorithm **waits until the signal degradation is significant enough** before switching.
- This delay avoids unnecessary handoffs when the mobile is near a cell boundary and signals fluctuate.
- It ensures that the handoff only occurs when the new cell's signal is clearly better, reducing the **ping-pong effect** (frequent back-and-forth switching).

Example

Imagine a mobile user moving near the boundary of Cell A and Cell B:

- Signal from Cell A drops slightly below the *warning threshold*.
- The system begins monitoring Cell B but does not hand off yet.
- If the user moves back toward Cell A, the call continues without interruption.
- Only if Cell A's signal falls below the *critical threshold* and Cell B's signal is stronger does the handoff occur.
- This delayed action prevents unnecessary switching and improves call stability.



Advantages

1. **Reduces unnecessary handoffs** caused by minor signal fluctuations.
2. **Improves call continuity** by avoiding premature switching.
3. **Optimizes network resources** by lowering signaling overhead.
4. **Enhances user experience** with fewer dropped calls and smoother mobility.

11. Discuss about queuing of handoffs.

A. Queuing of handoffs is a cellular network strategy to manage call drops by holding handoff (or handover) requests in a queue when no channels are available in the target cell. This prioritizes active calls, reducing user frustration from dropped connections, but requires careful tuning to avoid excessive queue lengths that can still block calls or delay them beyond acceptable limits. Key considerations include the trade-off between handoff failure probability and new call blocking probability, the size of the queue, and the use of "guard channels" which reserve a portion of channels specifically for handoffs.

Handoff Queuing Works

1. Handoff Request:

A mobile device, moving between cell coverage areas, initiates a handoff request.

2. Channel Unavailability:

If the target cell has no free channels, the handoff request is placed in a queue instead of being immediately blocked or dropped.

3. Queueing Mechanism:

The queued requests are then served in order (often first-come, first-served, or FCFS) as channels become available.

4. Call Completion:

The call is successfully handed off when a channel is free, or the call is terminated if the queue time expires.

Benefits

- Reduces Forced Termination:

By queuing handoff calls, the system prevents active calls from being prematurely disconnected due to lack of channels.

- Improves User Experience:

Users generally find a dropped call more frustrating than an inability to make a new one, making handoff queuing a priority for quality of service (QoS).

Key Considerations and Trade-offs

- Queue Length:

There is an optimal queue length; a very large queue can lead to long wait times and potentially exceed acceptable delay limits.

- New Call Blocking:

While handoff queuing prioritizes active calls, it might slightly increase the probability of new calls being blocked because fewer channels are available for them.

- Guard Channels:

Systems can use a combination of queuing and guard channels, where a set number of channels are reserved exclusively for handoffs, to further reduce handoff failures.

- Cellular Architectures:

In multi-tier or microcellular systems, where handoffs are more frequent, handoff queuing becomes even more critical for maintaining network performance.

- Dynamic Channel Allocation:

Efficient handoff queuing can be particularly beneficial with dynamic channel assignment schemes.

12. Define the dropped call rate and explain how it is related to the capacity and voice quality.

A. Definition of Dropped Call Rate (DCR)

- **Dropped Call Rate (DCR)** is the percentage of calls that are unintentionally terminated by the network after being successfully established.
- Formula:

$$DCR = \frac{\text{Number of dropped calls}}{\text{Total number of successfully established calls}} \times 100$$

- A dropped call occurs when the connection is lost due to poor signal, interference, handoff failure, or lack of resources—not because the user ended the call.

Relation to Capacity

1. High Traffic Load:

- When many users are connected, the network may run out of available

channels.

- If a handoff cannot be completed due to lack of free channels, the call is dropped, increasing DCR.

2. Resource Allocation:

- Networks often prioritize ongoing calls over new ones (e.g., queuing handoffs, guard channels).
- Efficient capacity management reduces DCR.

3. Congestion:

- Overloaded cells lead to higher dropped calls, showing a direct link between **capacity limits** and DCR.

Relation to Voice Quality

1. Signal Strength and Quality:

- Poor radio conditions (low RxLev, high interference) degrade voice quality.
- If conditions worsen beyond thresholds, the call drops, raising DCR.

2. Interference Management:

- High interference reduces clarity and increases the chance of dropped calls.

3. User Experience:

- A high DCR reflects poor voice quality and reliability, directly impacting customer satisfaction.
- Operators monitor DCR as a **Key Performance Indicator (KPI)** for service quality.